



FRONTEND — DETAILED BREAKDOWN

Tech Stack

- **React 18 + TypeScript**
 - **TailwindCSS** (styling)x
 - **D3.js** (custom/complex visualisations)
 - **Recharts** (standard charts)
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Screen-by-Screen Frontend Features

1. Brand Manager Input Form (FR-09)

The entry point. Six fields:

Field	Type
Product Name	Text input
Category	Dropdown
Target Persona	Dropdown (Fitness Millennial, Budget Parent, Urban Health Seeker, Senior Wellness User)
Primary Benefit Idea	Free-text (NLP-parsed on submission to extract claim signals)
Key Ingredient	Optional text input
Target Price Tier	Dropdown — Mass / Mid / Premium / Ultra-Premium (<i>mandatory — new addition</i>)

- Must be completable in **under 2 minutes**
 - On submission, free-text benefit idea is parsed by NLP to extract embedded claim signals before the scan begins
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2. Scan Progress UI

After form submission, the user sees a **real-time progress tracker**:

- Status polling every **3 seconds**

- Progress bar advances through pipeline stages: `pending` → `scanning` → `scoring` → `ai_processing` → `complete`
 - Each stage label is shown so the user understands what the system is doing
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3. Failure Dashboard (FR-10) ⚠️ — Shown *BEFORE* Recommendations

This is ValueForge's **most distinctive UX decision**. The document states: *"Show where you will crash before showing where to go."*

- Appears as a **full-screen mandatory card** after scan completion
 - Matches the user's idea against a library of **500+ failed product launches** using semantic similarity
 - Surfaces the **top 3 closest historical failure patterns**
 - Each failure card shows: positioning used, failure reason, lesson learned
 - **User must explicitly acknowledge before proceeding** to recommendations
 - Failure reason types displayed: `taste_mismatch`, `claim_not_believed`, `price_value_disconnect`, `persona_wrong`, `market_not_ready`, `distribution_failure`, `brand_permission_gap`
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4. Competitive Map — Bubble Chart (FR-11)

An interactive D3/Recharts bubble chart:

Axis/Property	Meaning
X-Axis	Tier-Adjusted CDS — left = fresh at this price tier, right = saturated
Y-Axis	Consumer Response Score (CRS) — top = strong predicted resonance
Bubble Size	Proportional to trend velocity (larger = faster growing)
Bubble Color	Green = True Whitespace, Yellow = Conditional, Orange = Brand Whitespace, Red = Contested
Brand Manager's Idea	Plotted as a distinct star marker for direct comparison

The Y-axis uses **CRS** (the 4-filter consumer response score), not raw persona score — this adds the consumer simulation layer to the visual.

5. Whitespace Grid — 3-Axis View (FR-12)

A grid that packs 3 dimensions of information into a 2D table:

- **Rows:** Consumer Need Categories (Energy, Recovery, Immunity, Taste, Convenience, Sustainability)
- **Columns:** Market Coverage at Target Price Tier (Underserved / Moderate / Saturated)
- **Cell Color:** Green = True Whitespace, Yellow = Conditional, Red = Contested, Grey = Brand Permission Gap
- **Cell Opacity** (the third dimension): Fully opaque = strong brand fit; Faded = poor brand fit (BPS-driven)

6. Value Proposition Cards (FR-13)

Top 3 claims ranked by FOS, each displayed as a rich card with **7 sections**:

Section	Content
A — Claim Headline	AI-generated positioning headline + supporting sub-claims
B — Opportunity Scores	FOS badge, Tier-CDS zone indicator, CRS breakdown (4 filter bars), BPS indicator, trend direction arrow
C — Ingredient Recommendation	1–2 hero ingredients scientifically aligned with the claim and persona driver (e.g., Recovery + Achievement driver → Collagen Peptides + Electrolyte Blend)
D — Format & Packaging	Physical product format + packaging direction (e.g., "Single-serve stick pack, matte finish, minimalist label for premium fitness persona")
E — Price Architecture	Recommended price band (e.g., ₹120–150 per serve)
F — First Mover Window	Estimated months before this claim territory saturates at the target tier
G — Channel Fit	Top 2 marketing channels most aligned with persona + claim type

This moves ValueForge beyond messaging — each card is a **complete product design brief**.

7. Misalignment Risk Flags (FR-14)

Displayed as **red warning tags** on any flagged claim:

- Triggers when: TierCDS > 60 OR CRS < 40 OR BPS < 40
 - Shows which specific dimension failed
 - Provides a suggested replacement claim from the True Whitespace set
 - Brand Permission (BPS < 40) is a **new first-class flag trigger** — not present in earlier versions
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8. Authentic Claim Territory View (FR-15)

A dedicated view showing the intersection of all three dimensions:

- Visualised as a **Venn diagram** or **3-axis radar chart**
 - Each claim is plotted against all three dimensions (Market, Consumer, Brand)
 - The overlap zone = Authentic Claim Territory — the only claims the brand can both find AND win
 - Output: ranked list of claims that sit inside this overlap zone
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9. Brand Brief Preview & Export (FR-16)

An in-app preview and export of the complete Brand Brief, containing 8 sections:

1. Product Overview
2. Failure Simulation Results
3. Top 3 Value Propositions with Product Design Directions
4. Claims to Avoid (with reasons)
5. Authentic Claim Territory Map
6. Trend Window & First Mover Deadlines
7. Competitive Landscape Summary
8. Risk Flags

Export formats: **PDF** (primary) and **DOCX** (secondary). Download link is valid for **15 minutes** via a signed S3 URL.